Sports Biomechanics

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/rspb20

Are sprint accelerations related to groin injuries? A biomechanical analysis of adolescent soccer players

Thomas Dupré & Wolfgang Potthast

To cite this article: Thomas Dupré & Wolfgang Potthast (19 Oct 2022): Are sprint accelerations related to groin injuries? A biomechanical analysis of adolescent soccer players, Sports Biomechanics, DOI: 10.1080/14763141.2022.2133740

To link to this article: https://doi.org/10.1080/14763141.2022.2133740

Published online: 19 Oct 2022.

Submit your article to this journal

Article views: 337

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=rspb20

SPORTS BIOMECHANICS
https://doi.org/10.1080/14763141.2022.2133740

Routledge
Taylor & Francis Group

Check for updates

Are sprint accelerations related to groin injuries? A biomechanical analysis of adolescent soccer players

Thomas Dupré[a,b] and Wolfgang Potthast[a]

[a]Institute of Biomechanics and Orthopedics, German Sport University Cologne, Cologne, Germany;
[b]Department of Exercise Science, Olympic Training Centre Rhineland-Palatinate/Saarland, Mainz, Germany

ABSTRACT

Groin injuries have one of the highest incidences in soccer and can be career threatening, especially for adolescents, due to their high recurrence rate. Quick accelerations have been connected to groin injuries along with kicking and change of directions. Purpose of this study was to examine the hip joint kinematics, kinetics and the muscle forces of adductor longus and gracilis during first ground contact of a linear sprint acceleration performed by adolescent soccer players. Twenty-two male participants were investigated with 3D motion capture and two force plates. Inverse dynamics were used to calculate the kinematics, kinetics and muscle forces. The kinematics show a constant extension during the stance phase and a quick transition from an abduction to an adduction movement at 90% stance, which coincides with the highest forces in adductor longus and gracilis. This indicates a high load on the adductor muscles due to eccentric contractions combined with high muscle forces in the adductors. Compared to previously investigated inside passing and change of direction movements, adductor muscle forces and angular velocities are higher in this study. Therefore, it is suggested that sprint accelerations are likely to be connected to the development of groin injuries in adolescent soccer players.

ARTICLE HISTORY
Received 16 May 2022
Accepted 4 October 2022

KEYWORDS
Groin injuries; soccer; adductors; adolescents; injury risk

Introduction

In soccer, a high number of injuries are groin related, with adductor injuries being the most common groin injury (GI) (Ekstrand, Hägglund, & Waldén, 2011; Hägglund, Waldén, & Ekstrand, 2013). Such injuries can cause a high amount of loss from training, with an average of 14 to 17 days lost (Ekstrand et al., 2011; Lewin, 1989). The risk for reinjuries is also increased after the initial GI (Hägglund et al., 2013), whereupon the subsequent injuries can cause an even higher training loss (Werner, Hägglund, Waldén, & Ekstrand, 2009). Therefore, GI have been attributed to cause a career-threatening 'vicious cycle of injury and reinjury' (Whittaker, Small, Maffey, & Emery, 2015) in some cases. While the scope of the problem, especially in soccer, has been investigated in numerous epidemiological studies (Ekstrand & Hilding, 1999; Hölmich, Thorborg,

CONTACT Thomas Dupré thomasdupre02@gmail.com
© 2022 International Society of Biomechanics in Sports

Dehlendorff, Krogsgaard, & Gluud, 2014; Orchard, 2015), the number of biomechanical studies investigating possible injury mechanisms is small.

It is thought that GI often occur due to repeated stress on the adductor muscles and pubic symphysis (Hiti, Stevens, Jamati, Garza, & Matheson, 2011). Previous studies have mostly investigated kicking and change of direction movements regarding their possible involvement in the development of GI (Brophy, Backus, Pansy, Lyman, & Williams, 2007; Dupré et al., 2020; Dupré & Potthast, 2020; Edwards, Brooke, & Cook, 2017; Franklyn-Miller et al., 2017). They have shown that both movements are likely to be involved in the development of GI. Both are defined by long periods of eccentric contractions in the adductor muscle group, which puts the muscles under high load and might lead to strain injuries (Charnock, Lewis, Garrett, & Queen, 2009; Dupré et al., 2018; Dupré, Tryba, & Potthast, 2021; LaStayo et al., 2003). In the case of changes of direction, this is intensified due to the fact that the highest muscle activation occurs at the end of the stance phase during the re-acceleration, often with an abducted hip joint and therefore elongated adductor muscles (Dupré et al., 2021). Inside passing on the other hand, leads to high muscle stress in the adductors and especially in the gracilis (Dupré et al., 2018), which is thought to be a potential injury risk considering the repetitive nature of inside passing (Garrett, 1990; Rahnama, Reilly, & Lees, 2002). While these are first insights into the possible origins of GI, it is thought that rapid accelerations also increase the risk for GI (Morelli & Weaver, 2005; Whittaker et al., 2015). Studies on sprint start biomechanics have so far mostly concentrated on understanding the performance determining kinematic and kinetic factors (Bezodis, Willwacher, & Salo, 2019; Brazil et al., 2016). However, there are no biomechanical studies investigating the role of sprinting and quick accelerations in the development of GI.

GI are also evident in adolescents (Raya-González et al., 2019) where the incidence of GI peaks in age groups close to the growth spurt (Light et al., 2021). The above-mentioned cycle of injury and reinjury can threaten the careers of young soccer players early on, as it reduces the chances of making it to the elite team (Larruskain et al., 2021). Little is known regarding the risk factors for GI in youth soccer players: A higher BMI is associated with developing groin pain in male players (Schoffl, Dooley, Miller, Miller, & Snodgrass, 2021) as well as being an early-maturing-player (Le Gall, Carling, & Reilly, 2007), but no further risk factors have been identified. Other studies have shown that the adolescent growth spurt increases the muscular demands in adolescent soccer players (Dupré et al., 2020; Light et al., 2022) which might influence the development of overuse injuries during this period. The growth spurt also results in changes to the kinematic profiles at the hip joint in both inside passing (Dupré et al., 2020) and 90° change of direction movements (Dupré & Potthast, 2020). Such insights provide valuable starting points for further studies into injury prevention. However, the role of quick accelerations regarding the development of GI in adolescents has not been studied so far. Previous research has investigated how the sprint biomechanics change during childhood growth: Nagahara et al. (2018) showed that the development of spatio-temporal parameters plateaus between the age of nine and twelve years, compared to the rapid increase which is seen before and after this time span. This effect is probably caused by the development of the central nervous system (Wdowski, Noon, Mundy, Gittoes, & Duncan, 2020) and could also be connected to the so-called adolescent awkwardness (Philippaerts et al., 2006). But little is known about the hip joint kinematics and kinetics

of adolescents in general, and even less on adolescent soccer players. Further knowledge is needed, to understand the injury mechanisms behind GI, and to be able to reduce the injury incidence among adolescents. The first step towards this goal is the identification of movements which might contribute to the development of GI due to their kinematic and kinetic profiles.

Therefore, the purpose of this study was to investigate the hip joint kinematics, kinetics and adductor muscle forces of sprint acceleration (SA) in a group of soccer players that were close to the adolescent growth spurt. It is hypothesised that the hip joint kinematics and kinetics of SA show similar characteristics as movements previously associated with GI, which will give insights into possible injury mechanisms of GI.

Methods

Design

This study was based on a cross-sectional design to investigate hip joint kinematics, kinetics and muscle forces in the adductor muscle group of adolescent boys during sprint acceleration. 3D motion capture was used to capture kinematic data and ground reaction forces were measured with two force plates. Kinematic and kinetic parameters of the hip joint were calculated with an inverse dynamics approach. The following parameters were investigated: Joint angles, angular velocities and external moments of the hip joint in all three movement planes as well as the muscle forces of adductor longus and gracilis.

Participants

Twenty-two adolescent male soccer players (13.43 ± 0.74 years, 1.62 ± 0.1 m, 49.7 ± 8.9 kg), were recruited from local soccer clubs of the district league. Exclusion criteria were any acute or chronic injuries to the lower extremity as well as injuries of the musculoskeletal system of the upper body that might hinder normal physical performance. Training load of the participants was three to four trainings per week with the addition of one match per weekend. Every participant and their parents or legal guardians gave their written consent before the start of the measurement. The study was designed according to the requirements of the Declaration of Helsinki and was approved by the German Sport University Cologne's ethics committee (Nr. 094/2018).

Procedure

Testing was performed in a motion analysis laboratory at the German Sport University Cologne. Kinematic data was collected with a 13-camera setup (200 Hz, 12 MX-F40 and 1 MX-3, Vicon, Oxford, GB). To collect ground reaction force data, two force plates were embedded in the ground (90 cm × 60 cm, 1000 Hz, Kistler, Winterthur, CH). The floor of the laboratory and the force plates was covered with third generation artificial turf (Ligaturf RS Pro IICP, Polytan, Burgheim, GER) to provide a soccer-specific surface for the movements.

Before the start of the measurement, participants were informed of the procedure and could ask questions. Afterwards, bony reference points on the skin were marked,

which were later used to indicate the attachment points of the retro-reflective markers. Subsequently, anthropometric measurements, including segment lengths and circumferences, were taken. Measurements of the leg length and sitting height were also taken to perform an estimation of the participants maturity status as described by Mirwald, Baxter-Jones, Bailey, and Beunen (2002). This calculation results in an estimation of an adolescent's time to peak height velocity (PHV), which is the fastest increase in length during the adolescent growth spurt. The participants were then allowed 10 min of self-reliant warm-up. Twenty-eight retro-reflective markers were placed on the already marked bony landmarks of the lower body after warm-up and a standing reference was recorded before the measurement of the movements started.

All participants performed practice trials of SA before the measurement until they felt comfortable with the movement. They were instructed to perform SA as a maximum effort sprint start for at least 5 m. SA was performed from a standing start and five valid trials were collected with at least 1 min between trials. An onset of fatigue was not expected due to the short acceleration and small number of trials. There was no specific starting command and participants were allowed to start when they were ready. Participants had to stand on force plate 1 with the foot in front that was more comfortable to them (Figure 1). This was to ensure that their first step would touchdown on force plate 2 and could be analysed. Trials were valid if the touchdown of the first step was on force plate 2 with no double contact and no contact of the foot with the turf outside the force plate.

Data processing and modelling

Post-processing of the data was performed in Vicon Nexus 2.6 (Vicon, Oxford, UK). To perform inverse dynamics calculations on the kinematic and kinetic data, AnyBody Modelling System (Version 6.0, AnyBody Technology, Aalborg, DEN) was used with the Anatomical Landmark Scaled Model (Lund, Andersen, de Zee, & Rasmussen, 2015). The model was modified to incorporate a spherical knee joint with three degrees of freedom (Dupré, Dietzsch, Komnik, Potthast, & David, 2019). This model also uses three degrees of freedom at the hip joint and two at the ankle joint. Kinematic and kinetic data were filtered with a recursive fourth-order low-pass Butterworth filter and a cut-off frequency of 50 Hz. Net joint moments were modelled by AnyBody as external moments and later normalised to each participants' body mass. The segment masses and inertial properties are based on Klein Horsman et al., (2007) and scaled to the participant-specific anatomy (Lund et al., 2015). Static optimisation with a cubic cost-function was used to calculate muscle forces (Dupré et al., 2019). Data reduction was performed with GNU Octave 6.4.0 (Open Source by John W. Eaton). Length of the movement was normalised to force plate contact on force plate 2 for further analysis. The contact on force plate 2 was chosen to replicate an in-game situation, where quick accelerations out of a movement are more common than the initial standing start. A subject mean was calculated from the valid trials of each participant for every parameter which was subsequently used to calculate an overall mean over all participants. Means were checked for normality with the Shapiro-Wilk-Test and $\alpha = .05$.

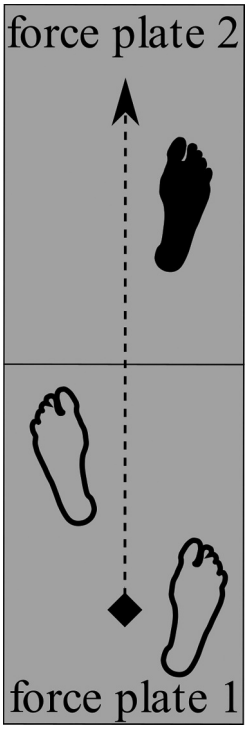

Figure 1. Schematic drawing showing the test setup. Subjects started from force plate 1 and made their first touchdown on force plate 2. The dashed arrow indicates the direction of movement.

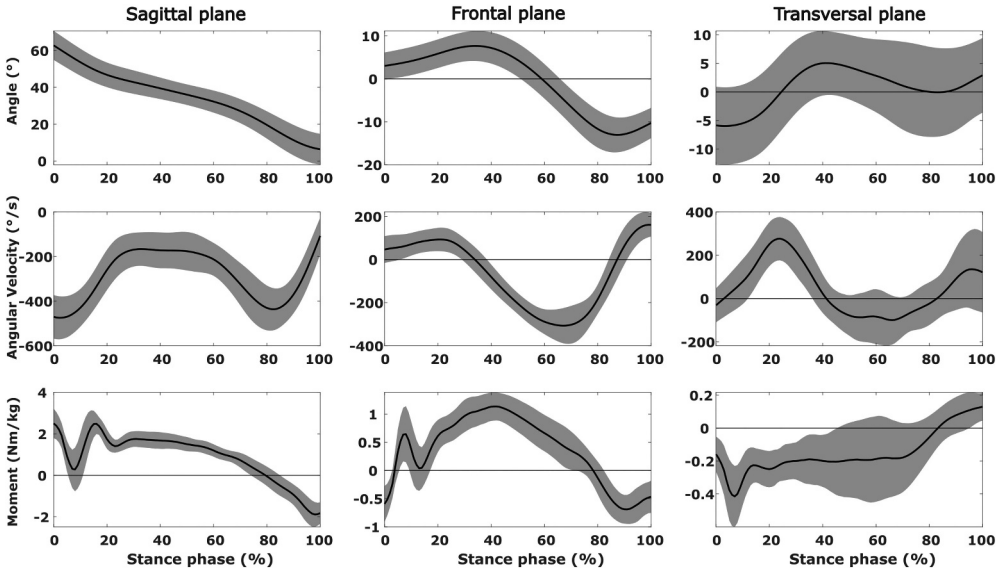

Figure 2. Average time series of the kinematics and kinetics from the hip joint of the 22 male participants during the stance phase of the initial step of a SA. The solid line represents the mean while the shaded area describes ± one standard deviation. Positive values indicate flexion, adduction and internal rotation in their respective planes of movement.

Results

Average time to PHV was 0.41 ± 1 years, resulting in an average age of 13.84 years old at the time of PHV. The hip joint kinematics and kinetics are presented as time series in Figure 2. Discrete maximum and minimum values of the kinematics and kinetics throughout the stance phase can be found in Table 1. Description of these time series will set a focus on the frontal plane kinematics and kinetics of the hip joint, since it is the most important in regard to a possible connection to GI. At touchdown, the hip is in an adducted position and is further adducted until it reaches its maximum at 33% of the stance phase. The hip is then abducted at an increasing angular velocity with maximum abduction velocity reached at 67% of the stance phase and maximum abduction angle reached at 88% of the stance phase. Before toe-off, the hip joint is adducted again and its maximum adduction velocity is reached on average at 70% of the stance phase. An external abduction moment is present in the frontal plane at touchdown which changes into an adduction moment at 5% of the stance phase. Afterwards, there is an adduction moment that reaches its maximum at 35% of the stance phase. The hip joint moment switches back to an abduction moment shortly before toe-off at 80% of the stance phase. On average, the maximum abduction moment is reached at 57% stance phase.

Figure 3 shows the time series of the adductor muscle forces during the stance phase of the first contact. The maximum muscle forces of gracilis and adductor longus can be found in Table 1. After touchdown, both muscles experience forces <20 N. Subsequently, the two muscles reduce their calculated force below 1 N (adductor longus at 13% and gracilis at 17% stance phase). Muscle forces stay <1 N until the last third of the stance phase where both show a rapid increase in their respective force towards toe-off. Both muscles reach their maximum force at 96% of the stance phase from where there is a slight decrease until toe-off.

Table 1. Maximum and minimum means of the kinematics and kinetics of the hip joint and maximum values of the muscle forces during ground contact of sprint acceleration. Values are presented as means ± standard deviations, calculated from the 22 adolescent participants' means. Positive values correspond to flexion, adduction and internal rotation in their respective plane of movement whereas negative values correspond to extension, abduction and external rotation. Means and standard deviations from non-normal distributed parameters are indicated by †.

Parameter	Maximum	Minimum
Angles (°)		
Sagittal	64.02 ± 6.78	5.15 ± 6.16
Frontal	8.19 ± 3.30†	−13.52 ± 4.08
Transversal	7.47 ± 5.31	−7.47 ± 6.36
Angular velocities (°s $^{-1}$)		
Sagittal	−58.11 ± 73.99	−520.22 ± 72.14
Frontal	187.62 ± 50.23†	−333.28 ± 70.12†
Transversal	367.32 ± 93.91	−221.39 ± 90.71
Joint moments (Nm kg $^{-1}$)		
Sagittal	3.07 ± 0.66†	−2.17 ± 0.41
Frontal	1.39 ± 0.28	−0.97 ± 0.23
Transversal	0.18 ± 0.05	−0.54 ± 0.22†
Muscle forces (N)		
Adductor longus	94.00 ± 23.39	
Gracilis	86.07 ± 20.63	

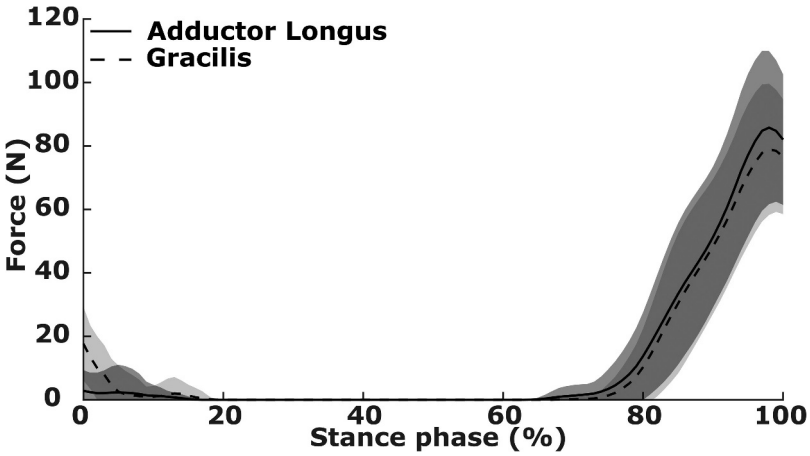

Figure 3. Average time series of the adductor muscle forces of the 22 male participants during the stance phase of the initial step of a SA. The solid line represents the mean of adductor longus and the dashed line represents gracilis. The shaded areas describe ± one standard deviation of each muscle.

Discussion and implications

The purpose of this study was to investigate the sprint start kinematics, kinetics and muscle forces of adolescent soccer players near PHV and whether they give insight into a possible injury mechanism for GI. To the authors knowledge, this is the first study to investigate these parameters in adolescent soccer players during SA. In the past, studies on adolescent SA kinematics and kinetics, have, similar to research on adult sprint biomechanics, tried to gain insights from a performance and/or growth-related stand point (Colyer, Nagahara, Takai, & Salo, 2020; Miyamoto, Takeshita, & Yanagiya, 2018; Wdowski et al., 2020), but have not investigated the origin of injuries in general and GI in particular.

Two mechanisms are thought to promote the development of adductor injuries: Eccentric contractions impose high forces on muscles and passive structures (LaStayo et al., 2003), thereby facilitating the development of GI in sports like hockey and soccer, which entail high amounts of movements with eccentric contractions (Brophy et al., 2007; Charnock et al., 2009; Dupré et al., 2018). Furthermore, quick transitions between flexion and extension are thought to place the adductor muscles at risk of strain injuries (Charnock et al., 2009). Recent studies on the development of GI in soccer athletes have found both mechanisms to be present during changes of directions and inside passing and concluded, that in combination with the high number of repetitions of these movements during training and matches, the risk for suffering adductor injuries is likely to be increased (Charnock et al., 2009; Dupré et al., 2018, 2021). Similar characteristics of the kinematics and kinetics are evident in the present study, especially in the frontal plane, where the adductor muscles do the majority of their work. The frontal plane hip angle performs a quick transition from adduction towards abduction during stance. Since first stance of SA has been reported to be 200 ms (Nagahara et al., 2018), this transition is comparable to transitions found in kicking, which are suggested to put the adductor longus under high stress (Serner, Mosler, Tol, Bahr, & Weir, 2018). Although the

maximum abduction angle is smaller compared to inside passing and change of direction movements, the maximum abduction velocity, responsible for the transition between adduction and abduction, of 333°s^{-1} is twice as high as during inside passing and three times as high as during a 90° change of direction movement of adolescents (Dupré et al., 2020; Dupré & Potthast, 2020). Towards the end of the stance phase, the abduction angle and velocity have to be reversed into adduction by the adductor muscles. Consequently, the highest adductor forces occur simultaneously with the external abduction moment during final stance (see Figures 2 and 3), effectively pulling the thigh back towards the centre of the body. The ensuing maximum muscle force is higher during first stance of SA than during change of direction manoeuvres or soccer inside passing (Dupré et al., 2020; Dupré & Potthast, 2020). In inside passes it has been suggested that repeated high muscle stress in the adductors and especially the gracilis, could lead to overuse injuries of these muscles and also their tendons, as the mechanical stress in the muscle multiplies towards the apophysis, due to its significantly smaller diameter (Dupré et al., 2018). Unfortunately, no data on the physiological cross-sectional area of adolescent adductor muscles could be found, making a calculation of the muscle stress impossible. It is, however, possible to assume that the muscle stress is higher in SA compared to inside passing, since the forces reported here are higher than those previously calculated for inside passing of adolescents (Dupré & Potthast, 2020). Therefore, it is evident that SA puts the adductor muscles repeatedly under high load due to the high adductor muscle forces and presumably high stress in the gracilis, which could impact the risk for GI.

When comparing the stance-phase kinematics and kinetics of SA with one reported for a 90° change of direction manoeuvre, there are several noteworthy similarities: In the sagittal plane, the hip is in a flexed position at touchdown and experiences a continuous extension until toe-off. Consequently, the angular velocities stay in the extension range throughout both movements, but show a sharp decrease in extension velocity towards the end of the stance phase (Dupré & Potthast, 2020). The hip joint moments of the sagittal and frontal plane appear to be also similar between 90° change of direction and SA. Both movements show a flexion moment at touchdown of 2 Nm kg^{-1}, which switches into an extension moment at around 80% of the stance phase. In the frontal plane, both movements start with an external abduction moment, which transitions into an adduction moment during mid stance before the ground contact ends with an abduction moment of comparable height between SA and 90° change of direction (Dupré & Potthast, 2020). Additionally, muscle forces are similar in appearance with almost no force during mid stance and the maximum muscle forces in the adductor longus and gracilis occurring before toe-off (Dupré & Potthast, 2020). Since it has been argued that GI are often a result of high loads and eccentric contractions, the similarities in the kinematic, kinetics and muscle forces between SA and change of direction manoeuvres, make it probable, that SA, in combination with inside passing and changes of direction, contribute to the high incidences of GI in adolescents and most likely adults.

The previous argument could also have implications for the assessment of change of direction manoeuvres, regarding the movement phase where the highest load to the groin area occurs: Previously, it has been argued that the injury risk is induced by high forces during mid stance of change of direction manoeuvres (Chaudhari, Jamison, McNally, Pan, & Schmitt, 2014). Newer studies, however, have shown that terminal stance of change of direction manoeuvres seems to be the phase where injuries are most likely to

occur due to the high muscle forces and eccentric contractions (Dupré & Potthast, 2020; Serner et al., 2018). Taking into account, that SA and change of direction manoeuvres show similar kinematics, kinetics and adductor muscle forces, the assumption can be made that it is not the change of direction which makes GI more likely during change of direction manoeuvres, but rather the phase of acceleration at the end of the movement. These findings interconnect with an increased GI incidence in ice hockey players (Emery & Meeuwisse, 2001). Ice hockey accelerations, similar to SA, require quick changes between adduction and abduction in the hip joint, while also relying on high abduction angles (Shell et al., 2017). Likely, this results in high muscle stress during eccentric adductor contractions, when performing such accelerations, which results in similar loads as in SA.

The calculated mean age at PHV was identical to Philippaerts et al. (2006) and therefore lay in between other results previously reported (Bult, Barendrecht, & Tak, 2018; Colyer et al., 2020). It has been argued that adolescent athletes are exposed to higher injury risks due to their quick growth during PHV and a reorganisation of the central nervous system (Colyer et al., 2021; Dupré et al., 2020; Philippaerts et al., 2006). In the present study, it is also evident that adolescent soccer players experience high loads on the adductor muscle complex during SA, due to the muscle forces and fast angular velocities in the frontal plane of the hip joint. Although it is unclear if this is specific to adolescents and/or soccer players, it suggests that GI prevention should be a consideration in the training regimen of these groups.

Interestingly, there is no evidence of a high incidence of GI in track and field athletes in general and sprinters specifically, although SA might induce critical loads to the groin area, given the results presented here. Soccer is a combination of a multitude of different movements like kicking, changes of directions and SA, with high muscular and structural demands on the groin area by each of these movements and it might be the combination of these tasks, that leads to an overload of structures in the groin area and subsequently an injury. Sprint athletes on the other hand, only perform linear accelerations and running during training and competition, while movements like kicking or changes of direction are not existent in their regimen. Therefore, they accumulate less loads on the groin area, making an overload and injury less likely. Sprint training also includes high amounts of specific strength training, particularly for the lower extremity which should provide a higher tolerance for high adductor muscle forces. However, this is hypothetical and should be investigated further because the present study did not include sprint athletes to investigate this argument specifically.

Limitations

To predict the time to PHV, the method by Mirwald et al. (2002) was used. This method is only able to predict the time to PHV through anthropometric measures while the gold standard would be to use radiography to determine the skeletal age. Although very accurate, this will expose the subjects to radiation and should only be used if absolutely necessary. Therefore, the indirect method by Mirwald was used in this study. Using static optimisation to calculate muscle forces is a limiting factor because the force calculation is based upon the net joint moments getting distributed to the muscles acting upon the joint. Therefore, any stabilising muscle forces acting on the joint are not represented by

the calculated muscle forces. This results in periods without any significant muscle forces being present in the investigated muscles as is evident in Figure 3. However, the particular model used in the present study has shown good accordance between modelled muscle activity and measured muscle activation (Dupré et al., 2019). With the use of markers for the 3D video analysis, it is possible that the participants experienced slight irritations that resulted in variations to their movements. Since the used markers were small and the participants had time to get accustomed to the markers before the measurement, possible effects on the results are deemed to have been minimal.

Conclusion

The present findings suggest that adolescent soccer players experience high loads to the adductor longus and gracilis, as well as the encompassing passive structures of the groin area, when performing maximum sprint accelerations. This is evident due to the high muscle forces and eccentric contractions that occur during this task. In combination with inside passes and changes of directions, which also put high loads on the groin area, it is likely that high adductor loads and high abduction angles found in sprint accelerations, are connected to the high incidences of groin injuries, present in adolescent soccer players. Future studies should try to quantify the risk associated with these movements in longitudinal studies. Electromyographic investigations into the adductor muscle activation during SA would also benefit the understanding of the muscle mechanics and possible injury mechanisms.

Acknowledgements

The authors wish to thank Corinne Kohlmann, Julian Tryba and Max Kreienkamp for their help during data acquisition and post processing. The authors also wish to thank Niklas Hagenhoff for his help in participant acquisition.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This study was supported by the family bureau of the German Sport University Cologne.

ORCID

Thomas Dupré http://orcid.org/0000-0003-0434-6095

References

Bezodis, N. E., Willwacher, S., & Salo, A. I. T. (2019). The biomechanics of the track and field sprint start: A narrative review. *Sports Medicine*, *49*(9), 1345–1364. https://doi.org/10.1007/s40279019-01138-1

Brazil, A., Exell, T., Wilson, C., Willwacher, S., Bezodis, I., & Irwin, G. (2016). Lower limb joint kinetics in the starting blocks and first stance in athletic sprinting. *Journal of Sports Sciences*, 1–7. https://doi.org/10.1080/02640414.2016.1227465

Brophy, R. H., Backus, S. I., Pansy, B. S., Lyman, S., & Williams, R. J. (2007). Lower extremity muscle activation and alignment during the soccer instep and side-foot kicks. *The Journal of Orthopaedic and Sports Physical Therapy*, *37*(5), 260–268. https://doi.org/10.2519/jospt.2007.2255

Bult, H. J., Barendrecht, M., & Tak, I. J. R. (2018). Injury risk and injury burden are related to age group and peak height velocity among talented male youth soccer players. *Orthopaedic Journal of Sports Medicine*, *6*(12), 1–10. https://doi.org/10.1177/2325967118811042

Charnock, B. L., Lewis, C. L., Garrett, W. E., & Queen, R. M. (2009). Adductor longus mechanics during the maximal effort soccer kick. *Sports Biomechanics*, *8*(3), 223–234. https://doi.org/10.1080/14763140903229500

Chaudhari, A. M. W., Jamison, S. T., McNally, M. P., Pan, X., & Schmitt, L. C. (2014). Hip adductor activations during run-to-cut manoeuvres in compression shorts: Implications for return to sport after groin injury. *Journal of Sports Sciences*, *32*(14), 1333–1340. https://doi.org/10.1080/02640414.2014.889849

Colyer, S., Miles, J. J., Crump, F. J., Hall-Martinez, J. P., Little, G. S., Mallabone, J., & Cumming, S. P. (2021). Association between biological maturation and anterior cruciate ligament injury risk factors during cutting. *The Journal of Sports Medicine and Physical Fitness*, *62*(8). https://doi.org/10.23736/S0022-4707.21.12590-3

Colyer, S., Nagahara, R., Takai, Y., & Salo, A. (2020). The effect of biological maturity status on ground reaction force production during sprinting. *Scandinavian Journal of Medicine & Science in Sports*, *30*(8), 1387–1397. https://doi.org/10.1111/sms.13680

Dupré, T., Dietzsch, M., Komnik, I., Potthast, W., & David, S. (2019). Agreement of measured and calculated muscle activity during highly dynamic movements modelled with a spherical knee joint. *Journal of Biomechanics*, *84*, 73–80. https://doi.org/10.1016/j.jbiomech.2018.12.013

Dupré, T., Funken, J., Müller, R., Mortensen, K. R. L., Lysdal, F. G., Braun, M., & Potthast, W. (2018). Does inside passing contribute to the high incidence of groin injuries in soccer? a biomechanical analysis. *Journal of Sports Sciences*, *36*(16), 1827–1835. https://doi.org/10.1080/02640414.2017.1423193

Dupré, T., Lysdal, F. G., Funken, J., Mortensen, K. R. L., Müller, R., Mayer, J., & Potthast, W. (2020). Groin injuries in soccer: Investigating the effect of age on adductor muscle forces. *Medicine and Science in Sports and Exercise*, *52*(6), 1330–1337. https://doi.org/10.1249/MSS.0000000000002243

Dupré, T., & Potthast, W. (2020). Groin injury risk of pubertal soccer players increases during peak height velocity due to changes in movement techniques. *Journal of Sports Sciences*, *38*(23), 2661–2669. https://doi.org/10.1080/02640414.2020.1794769

Dupré, T., Tryba, J., & Potthast, W. (2021). Muscle activity of cutting manoeuvres and soccer inside passing suggests an increased groin injury risk during these movements. *Scientific Reports*, *11*(1). https://doi.org/10.1038/s41598-021-86666-5

Edwards, S., Brooke, H. C., & Cook, J. L. (2017). Distinct cut task strategy in Australian football players with a history of groin pain. *Physical Therapy in Sport: Official Journal of the Association of Chartered Physiotherapists in Sports Medicine*, *23*, 58–66. https://doi.org/10.1016/j.ptsp.2016.07.005

Ekstrand, J., Hägglund, M., & Waldén, M. (2011). Injury incidence and injury patterns in professional football: The UEFA injury study. *British Journal of Sports Medicine*, *45*(7), 553–558. https://doi.org/10.1136/bjsm.2009.060582

Ekstrand, J., & Hilding, J. (1999). The incidence and differential diagnosis of acute groin injuries in male soccer players. *Scandinavian Journal of Medicine & Science in Sports*, *9*(2), 98–103. https://doi.org/10.1111/j.1600-0838.1999.tb00216.x

Emery, C. A., & Meeuwisse, W. H. (2001, Sep). Risk factors for groin injuries in hockey. *Medicine & Science in Sports & Exercise*, *33*(9), 1423–1433. https://doi.org/10.1097/00005768-20010900000002

Franklyn-Miller, A., Richter, C., King, E., Gore, S., Moran, K., Strike, S., & Falvey, E. C. (2017). Athletic groin pain (part 2): A prospective cohort study on the biomechanical evaluation of change of direction identifies three clusters of movement patterns. *British Journal of Sports Medicine*, *51*(5), 460–468. https://doi.org/10.1136/bjsports-2016-096050

Garrett, W. E. (1990). Muscle strain injuries: Clinical and basic aspects. *Medicine and Science in Sports and Exercise*, *22*(4), 436–443. https://doi.org/10.1249/00005768-199008000-00003

Hägglund, M., Waldén, M., & Ekstrand, J. (2013). Risk factors for lower extremity muscle injury in professional soccer: The UEFA injury study. *The American Journal of Sports Medicine*, *41*(2), 327–335. https://doi.org/10.1177/0363546512470634

Hiti, C. J., Stevens, K. J., Jamati, M. K., Garza, D., & Matheson, G. O. (2011). Athletic osteitis pubis. *Sports Medicine*, *41*(5), 361–376. https://doi.org/10.2165/11586820-000000000-00000

Hölmich, P., Thorborg, K., Dehlendorff, C., Krogsgaard, K., & Gluud, C. (2014). Incidence and clinical presentation of groin injuries in sub-elite male soccer. *British Journal of Sports Medicine*, *48*(16), 1245–1250. https://doi.org/10.1136/bjsports-2013-092627

Klein Horsman, M. D., Koopman, H. F. J. M., van der Helm, F. C. T., Poliacu Pros´e, L., & Veeger, H. E. J. (2007). Morphological muscle and joint parameters for musculoskeletal modelling of the lower extremity. *Clinical Biomechanics*, *22*(2), 239–247. https://doi.org/10.1016/j.clinbiomech.2006.10.003

Larruskain, J., Lekue, J. A., Martin-Garetxana, I., Barrio, I., McCall, A., & Gil, S. M. (2021, Aug). Injuries are negatively associated with player progression in an elite football academy. *Science and Medicine in Football*, 1–10. https://doi.org/10.1080/24733938.2021.1943756

LaStayo, P. C., Woolf, J. M., Lewek, M. D., Snyder Mackler, L., Reich, T., & Lindstedt, S. L. (2003). Eccentric muscle contractions: Their contribution to injury, prevention, rehabilitation, and sport. *Journal of Orthopaedic & Sports Physical Therapy*, *33*(10), 557–571. https://doi.org/10.2519/jospt.2003.33.10.557

Le Gall, F., Carling, C., & Reilly, T. (2007). Biological maturity and injury in elite youth football. *Scandinavian Journal of Medicine & Science in Sports*, *17*(5), 564–572. https://doi.org/10.1111/j.16000838.2006.00594.x

Lewin, G. (1989). The incidence of injury in an English professional soccer club during one competitive season. *Physiotherapy*, *75*(10), 601–605. https://doi.org/10.1016/S0031-9406(10)62366-8

Light, N., Johnson, A., Williams, S., Smith, N., Hale, B., & Thorborg, K. (2021, Mar). Injuries in youth football and the relationship to player maturation: An analysis of time-loss injuries during four seasons in an English elite male football academy. *Scandinavian Journal of Medicine & Science in Sports*, *31*(6), 1324–1334. https://doi.org/10.1111/sms.13933

Light, N., Thorborg, K., Krommes, K., Nielsen, M. F., Thornton, K. B., Hölmich, P., & Ishøi, L. (2022) Rapid spike in hip adduction strength in early adolescent footballers: A study of 125 elite male players from youth to senior. *International Journal of Sports Physiology and Performance*, *17*(9), 1–8. https://doi.org/10.1123/ijspp.2022-0025.

Lund, M. E., Andersen, M. S., de Zee, M., & Rasmussen, J. (2015). Scaling of musculoskeletal models from static and dynamic trials. *International Biomechanics*, *2*(1), 1–11. https://doi.org/10.1080/23335432.2014.993706

Mirwald, R. L., Baxter-Jones, A. D. G., Bailey, D. A., & Beunen, G. P. (2002). An assessment of maturity from anthropometric measurements. *Medicine and Science in Sports and Exercise*, *34*(4), 689–694. https://doi.org/10.1097/00005768-200204000-00020

Miyamoto, A., Takeshita, T., Yanagiya, T., & Nordez, A. (2018). Differences in sprinting performance and kinematics between preadolescent boys who are fore/mid and rear foot strikers. *Plos One*, *13*(10), e0205906. https://doi.org/10.1371/journal.pone.0205906

Morelli, V., & Weaver, V. (2005, March). Groin injuries and groin pain in athletes: Part 1. *Primary Care: Clinics in Office Practice*, *32*(1), 163–183. https://doi.org/10.1016/j.pop.2004.11.011

Nagahara, R., Takai, Y., Haramura, M., Mizutani, M., Matsuo, A., Kanehisa, H., & Fukunaga, T. (2018). Age-related differences in spatiotemporal variables and ground reaction forces during sprinting in boys. *Pediatric Exercise Science*, *30*(3), 335–344. https://doi.org/10.1123/pes.2017-0058

Orchard, J. W. (2015, June). Men at higher risk of groin injuries in elite team sports: A systematic review. *British Journal of Sports Medicine*, *49*(12), 798–802. https://doi.org/10.1136/bjsports-2014-094272

Philippaerts, R. M., Vaeyens, R., Janssens, M., Van Renterghem, B., Matthys, D., Craen, R., & Malina, R. M. (2006). The relationship between peak height velocity and physical performance in youth soccer players. *Journal of Sports Sciences*, *24*(3), 221–230. https://doi.org/10.1080/02640410500189371

Rahnama, N., Reilly, T., & Lees, A. (2002). Injury risk associated with playing actions during competitive soccer. *British Journal of Sports Medicine*, *36*(5), 354–359. https://doi.org/10.1136/bjsm.36.5.354

Raya-González, J., Suárez-Arrones, L., Navandar, A., Balsalobre-Fern´andez, C., & S´aez de Villarreal, E. (2019). Effects of a prophylactic knee sleeve on anterior cruciate ligament loading during sport-specific movements. *Journal of Sport Rehabilitation*, *28*(1), 1–7. https://doi.org/10.1123/jsr.2019-0113

Schoffl, J., Dooley, K., Miller, P., Miller, J., & Snodgrass, S. J. (2021, December). Factors associated with hip and groin pain in elite youth football players: A cohort study. *Sports Medicine - Open*, *7*, 97. https://doi.org/10.1186/s40798-021-00392-w

Serner, A., Mosler, A. B., Tol, J. L., Bahr, R., & Weir, A. (2018). Mechanisms of acute adductor longus injuries in male football players: A systematic visual video analysis. *British Journal of Sports Medicine*, *53*(3), 158–164. https://doi.org/10.1136/bjsports-2018-099246

Shell, J. R., Robbins, S. M. K., Dixon, P. C., Renaud, P. J., Turcotte, R. A., Wu, T., & Pearsall, D. J. (2017, May). Skating start propulsion: Three-dimensional kinematic analysis of elite male and female ice hockey players. *Sports Biomechanics*, *16*(3), 313–324. https://doi.org/10.1080/14763141.2017.1306095

Wdowski, M. M., Noon, M., Mundy, P. D., Gittoes, M. J. R., & Duncan, M. J. (2020). The kinematic and kinetic development of sprinting and countermovement jump performance in boys. *Frontiers in Bioengineering and Biotechnology*, *8*. https://doi.org/10.3389/fbioe.2020.547075

Werner, J., Hägglund, M., Waldén, M., & Ekstrand, J. (2009). UEFA injury study: A prospective study of hip and groin injuries in professional football over seven consecutive seasons. *British Journal of Sports Medicine*, *43*(13), 1036–1040. https://doi.org/10.1136/bjsm.2009.066944

Whittaker, J. L., Small, C., Maffey, L., & Emery, C. A. (2015). Risk factors for groin injury in sport: An updated systematic review. *British Journal of Sports Medicine*, *49*(12), 803–809. https://doi.org/10.1136/bjsports-2014-094287